Internship Progress Report

CamGeo Project — Preparation, Photo-interpretation, and Data Quality

Host organization:	Novalitix
Intern:	Fofack Henri
Project:	CamGeo land cover mapping in Cameroon
Target milestone:	Gate G2 — GitHub issue #2 — release v0.3.0
Pilot area:	Southern Region (cocoa basin)
Period:	*September 21, 2026 – October 6, 2026*
Supervisors:	*BRELL SANWOUO & LEONEL*

1. Objectives Recap

The internship aims to build, check, and document the reference sample dataset needed to train the land cover mapping models for the CamGeo project. The main deliverable is a labeled and reviewed sample dataset of at least 350 points over the Southern Region, compliant with the `samples/README.md` schema, and directly usable by scikit-learn and the Google Earth Engine (GEE) pipeline.

This work corresponds to the **Gate G2** milestone (GitHub issue #2, release v0.3.0). It conditions the next stages of the project, led by interns 2 and 3 (classifier training and accuracy assessment).

2. Work Completed

2.1 GEE Environment Setup

Registration and validation of the Google Earth Engine account. Creation of the associated Google Cloud project and initial hands-on work in the Code Editor.

2.2 Sample Point Generation

Execution of the `gee/sampling/generate_sample_points.js` script over the Southern Region. The script generates a reproducible random draw (fixed seed) of points in the pilot area.

2.3 Data Export and Preparation

Export of the CSV file from GEE to Google Drive, followed by processing with a Python script in Google Colab:

- ☞ **renaming and normalizing columns;**
- ☞ **adding the 14 columns of the CamGeo schema;**
- ☞ **verifying WGS84 coordinates (EPSG:4326), identifier uniqueness, absence of duplicates, and missing values;**
- ☞ **producing two files:**
  - ☞ `samples_sud_v01_ceo.csv` — for import into CEO;
  - ☞ `samples_sud_v01.csv` — final CamGeo deliverable.

2.4 Collect Earth Online Project Setup

Creation of the `CamGeo - Sud - v01` project with:

- ☞ **four collection questions:** `class_code`, `class_name`, `confidence`, `notes`;
- ☞ **imagery sources:** Mapbox Satellite, Sentinel-2, Landsat 8 & 9, Dynamic World, OpenStreetMap;
- ☞ **100 m square plot (1 ha), one sample per plot;**
- ☞ **double review and quality audit planned in the protocol.**

The figure below shows the points generated in the Southern Region.

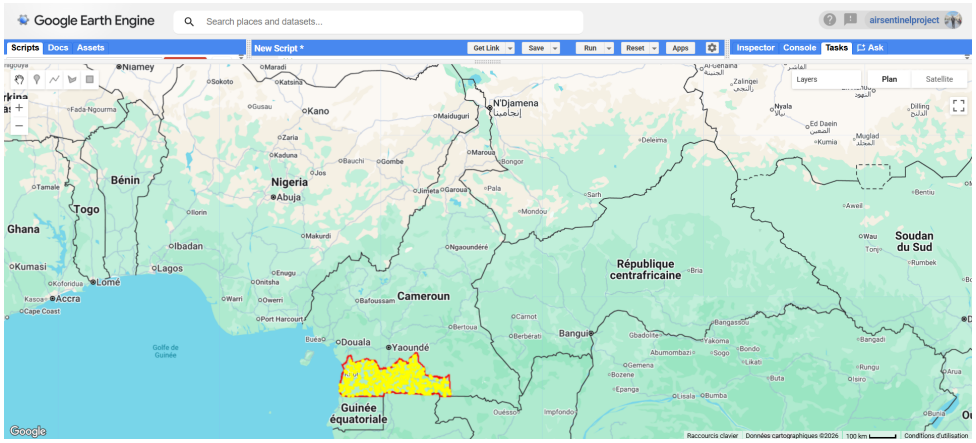


Figure 1: The 350 generated points.

2.5 Annotation Campaign

The annotation campaign is now well underway: **150 out of 349 points have been labeled** using the 10-class CamGeo legend. This represents more than half of the pending workload and confirms that the annotation method is stable and reproducible. The remaining points are being processed following the same protocol, with systematic use of Mapbox Satellite, Sentinel-2 and Landsat imagery, and careful documentation of ambiguous cases in the `notes` field.

3. Progress Status

Step	Done	Total
Points generated (GEE)	349	349
Points annotated (CEO)	150	349
Points reviewed (double review)	0	349
Random quality audit (10%)	0	35
GEE predictor extraction	not started	
Delivery of `samples_sud_v01.csv`	not started	

Table 1: Progress dashboard

Overall progress on the annotation phase is approximately **43%**.

4. Challenges Encountered and Proposed Solutions

- ☞ **Distinguishing dense forest (class 1) from shaded agroforestry (class 3).** These two classes have similar visual signatures. Solution: rely on landscape context (proximity to villages and roads), the presence of emergent trees, and canopy texture, in accordance with the annotation guide. This remains the most frequent ambiguity across the 150 points already processed.
- ☞ **Ambiguous cases (class boundaries, cloudy images).** Solution: use the `confidence: low` status and precisely document the nature of the doubt in the `notes` field. These cases are flagged for the double review phase.
- ☞ **Getting started with the CEO interface.** It took us 4 hours before project creation worked, after 6 unsuccessful attempts.
- ☞ **Time management.** With 150 points annotated, the pace is now stabilized, but the remaining 199 points must be completed while leaving sufficient time for the double review and the 10% quality audit.

5. Next Steps

1. Continue annotation until all 349 points are covered (approximately 199 points remaining).
2. Set up systematic double review (`review_status: reviewed`).
3. Perform the random 10% quality audit (approximately 35 points).
4. Extract geospatial predictors via GEE: Sentinel-2 bands, NDVI, NDWI, NDBI, Sentinel-1 SAR backscatter ratio (VV, VH, VV/VH), SRTM elevation, slope, aspect, and CHIRPS rainfall accumulations.
5. Document class counts and analyze imbalances specific to the Southern cocoa basin.
6. Deliver `samples_sud_v01.csv`, the GeoParquet version, the quality control report, and the methodological note, for the closure of the Gate G2 milestone.

6. Conclusion

The technical foundations of the internship are in place and the annotation campaign has reached a significant milestone with **150 out of 349 points labeled**. The annotation method is now stable, and the main remaining challenge is to maintain quality while scaling up. The coming weeks will focus on completing the remaining annotations, setting up the double review process, running the 10% quality audit, and preparing the extraction of geospatial predictors, in order to deliver a sample dataset fully compliant with the Gate G2 milestone.